# Mingling Yang

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## EDUCATION

**University of Southern California, Los Angeles, CA** Aug 2026 - Present
*Ph.D. in Epidemiology, Environmental Health Track*; Keck School of Medicine; Advisor: Vaia Lida Chatzi

**Johns Hopkins University, Baltimore, MD** Sep 2021 - May 2023
*M.S.E in Applied Mathematics and Statistics*; Whiting School of Engineering

- Relevant Courses: Mathematical Biology, Computational Molecular Medicine, Bayesian Statistics, Mathematical Image Analysis, Time Series Analysis, Probability Theory, Data Science, Machine Learning, Numerical Analysis
- Certifications & Training: Biostatistics in Public Health Specialization and Epidemiology in Public Health Practice Specialization (Johns Hopkins University); Causal Inference (Columbia University); Study Designs in Epidemiology (Imperial College London)

**University of South Carolina, Columbia, SC** Jan 2017 - Dec 2020
*B.S. in Statistics*, GPA: 3.86/4.00; McCausland College of Arts and Sciences
*Minor in Risk Management and Insurance*, GPA: 4.00/4.00

- Relevant Courses: Statistical Methods I&II&III, Statistical Inference, Calculus I&II&III, Linear Algebra, Mathematical Statistics, Algorithm Design in Java, SAS Language Programming, Stochastic Processes, Multivariate Statistics and Data Mining, Statistical Computing

## HONORS AND AWARDS

- Fully Funded Doctoral Support Package, University of Southern California 2026
- Departmental Tuition Fellowship, Johns Hopkins University 2021 - 2023
- Nonresident Scholarship, University of South Carolina
- Achievement Scholarship, University of South Carolina Fall 2020
- Dean's Honor List, University of South Carolina 2017 - 2020
- President's Honor List, University of South Carolina 2017 - 2020

## PUBLICATIONS AND CONFERENCES

### Published and Accepted Journal Articles:

1. Checkley W, **Mingling Yang**, Robertson NM, Sharma AK, Chandyo RK, Shrestha L, Das SK, Kirenga B, Alupo P, Gianella G, Siddharthan T, Pollard SL, Quaderi S, Rykiel N, Flores-Flores O, Hurst JR, Wise RA; Global Excellence in COPD outcomes (GECo) study investigators. Population-based Screening for Chronic Obstructive Pulmonary Disease Using the St. George's Respiratory Questionnaire in Resource-limited Settings. ***American Journal of Respiratory and Critical Care Medicine***. 2025 May; 211(5):779-788. doi: 10.1164/rccm.202409-1862OC. PMID: 40068167; PMCID: PMC12091018. (Featured in editorial)
2. Reuland CJ, Nicolaou L, Black MM, **Mingling Yang**, McCord R, Alvarado M, Arana F, Sanchez PS, Williams KN, Hossen S, Chiang M, Hartinger SM, Checkley W. Effects of Cooking with Liquefied Petroleum Gas or Biomass Fuels on Neurodevelopmental Outcomes in Preschool-Age Children Living in Puno, Peru. ***Environmental Health Perspectives***. 2025 Jun;133(6):67004. doi: 10.1289/EHP15500. Epub 2025 Jun 5. PMID: 40354572; PMCID: PMC12142655.
3. Robertson NM*, Sharma AK*, **Mingling Yang**, Das SK, Siddharthan T, Pollard SL, Rykiel NA, Alupo P, Flores-Flores O, Kirenga B, Chandyo RK, Quaderi SA, Shrestha L, Wise RA, Hurst JR, Checkley W; Global Excellence in COPD outcomes (GECo) Study Investigators. A Multi-country Cohort Study Evaluating the Prevalence, Risk Factors, Lung Function and Clinical Outcomes of Chronic Bronchitis in Low- and Middle-income Countries. ***European Respiratory Journal***. 2026 Mar; 67(3):2501435. doi: 10.1183/13993003.01435-2025. PMID: 41381225.
4. Nicolaou L, Reuland CJ, **Mingling Yang**, Williams KN, Hartinger SM, Chiang M, Checkley W. Long-Term Effects of Cooking with Liquefied Petroleum Gas or Biomass on Linear Growth Trajectories from Birth to the Pre-School Years in Puno, Peru: A Prospective Cohort Study. ***The Lancet Regional Health – Americas***. 2026 Mar; 55:101382. doi:10.1016/j.lana.2026.101382.
5. Checkley W, Robertson NM, **Mingling Yang**, Nicolaou L, Wise RA, Sharma AK, Chandyo RK, Shrestha LP, Das SK, Kirenga B, Alupo P, Gianella G, Siddharthan T, Pollard SL, Quaderi S, Rykiel N, Flores-Flores O, Hurst JR; Global Excellence in COPD outcomes (GECo) investigators. Prevalence of Chronic Obstructive Pulmonary

Disease and Population Attributable Risk across Three Resource-Limited Settings: The Global Excellence in COPD Outcomes (GECo) Study. ***The Lancet Global Health***. (Accepted)

6. Robertson N, Sharma A, **Mingling Yang**, Das S, Siddharthan T, Pollard S, Rykiel N, Alupo P, Flores-Flores O, Kirenga B, Chandyo R, Quaderi S, Shrestha L, Wise R, Hurst J, Checkley W. Reframing bronchodilator responsiveness using Z-scores: Multi-country evaluation of Z-score–based criteria. ***European Respiratory Journal***. (Accepted)

**Submitted and In-Preparation Manuscripts:**

1. Emetu SA, **Mingling Yang**, Nicolaou L, Kephart JL, Fandiño-Del-Río M, Wise RA, Williams KN, Steenland K, Koehler K, Checkley W; CHAP Trial Investigators. Effects of Cooking with Liquefied Petroleum Gas or Biomass on Lung Function in Adult Peruvian Women: A Randomized Controlled Trial. ***Thorax***. (Returned for formatting corrections; resubmission pending).
2. Building a high-resolution PM2.5 map for Bhaktapur, Nepal: a dynamic spatiotemporal calibration model for low-cost sensors. (In preparation)
3. Williams K*, **Mingling Yang**, Kephart JL, Mallikarachchi M, Fandiño del Rio M, Koehler K, Harvey SA, Checkley W. Social networks and clean-fuel-use behaviors in rural Peru: a social-network analysis of the CHAP trial. (In preparation)
4. Wells M*, Tarazona-Meza C*, **Mingling Yang**, Williams K, Nicolaou L, Steenland NK, Checkley W. Vitamin A and omega-3 fatty acid deficiencies and eye health among women in the HAPIN cohort. (Submitted)

**Abstracts and Presentations:**

1. **Mingling Yang**, N.M. Robertson, A.K. Sharma, R.K. Chandyo, L. Shrestha, S.K. Das, B. Kirenga, P. Alupo, G.E. Gianella, T. Siddharthan, S.L. Pollard, S. Quaderi, N. Rykiel, O. Flores-Flores, J.R. Hurst, R.A. Wise, and W. Checkley. Screening for Chronic Obstructive Pulmonary Disease (COPD) Using the St. George's Respiratory Questionnaire. *Am J Respir Crit Care Med.* 2025;211:A1247. doi: https://doi.org/10.1164/ajrccm.2025.211.Abstracts.A1247. Presented by Mingling Yang at: *American Thoracic Society International Conference*; 2025 May 18; San Francisco, CA, USA.
2. Robertson N*, **Mingling Yang**, et al. Chronic Bronchitis Prevalence, Risk Factors, and Outcomes in a Multinational Cohort. *Eur Respir J.* 2025;66:Suppl.69, PA463. doi: https://doi.org/10.1183/13993003.congress-2025.PA463. Presented by N. Robertson at: *European Respiratory Society International Congress*; 2025 Oct 1; Amsterdam, Netherlands.
3. Wells M, Nicolaou L, **Mingling Yang**, Williams KN, Checkley W. The Burden of Preventable Eye Disease in Rural Peru: Clinical Findings from the HAPIN Eye Substudy. Presented by M. Wells at: *Women in Ophthalmology Summer Symposium*; 2025 Aug 12; Amelia Island, FL, USA.

## RESEARCH EXPERIENCE

**Keck School of Medicine, University of Southern California**
*Research Assistant |Advisor: Vaia Lida Chatzi* Aug 16 – Dec 31, 2026

**Johns Hopkins University School of Medicine**
*Senior Research Data Analyst |Medicine, Epidemiology & Biostatistics* Sep 2023 – May 3, 2026

- **Household Air Pollution Intervention Network (HAPIN) Trial** Guatemala, India, Peru, Rwanda
  - Contributed to data visualization and replication of selected analyses in R, preparing an R Markdown report to document the analytical workflow and summarize the results.
  - **Output:** Contributed to "Exposure to Household Air Pollution From Biomass Cooking and Severe Pneumonia in Infants." **JAMA Network Open.** 2025; 8(10): e2538721.
- **Alliance for a Healthier World (AHW) - Ambient Air Pollution Project** Nepal & Peru
  - Conducted the first round of data analysis for Nepal and contributed to selected analyses for Peru using four years of ambient monitoring network and E-Sampler data ($n \approx 75{,}000{,}000$), with guidance from Laura Nicolaou and William Checkley.
  - Conducted preliminary calibration modeling for co-located PurpleAir low-cost $PM_{2.5}$ sensors against MetOne E-Sampler reference monitors using machine-learning algorithms (random forest, XGBoost), with model development remaining incomplete; performed in-field sensor co-location, deployment, and reference-grade calibration data collection.
  - Worked on preliminary dynamic spatiotemporal models incorporating meteorological covariates and geostatistical methods (Kriging) for hourly $PM_{2.5}$ mapping and examination of seasonal variation and urban-rural differences.
  - **Output:** Manuscript in preparation titled "Building a High-Resolution PM2.5 Map for Bhaktapur, Nepal: A Dynamic Spatiotemporal Calibration Model for Low-Cost Sensors."

- **HAPIN - Longitudinal Study of Early Child Growth** Puno, Peru
  - Curated study data; replicated and validated longitudinal mixed-effects analyses of length/height-for-age z-score (LAZ/HAZ) trajectories and exposure-response analyses of HAZ at ages 2–4 years in the HAPIN follow-up study (n=683).
  - Implemented reproducible R Markdown workflows for data cleaning, visualization, and modeling to prepare the published online supplement.
  - Contributed to manuscript review and editing.
  - **Output:** Co-author; Published in **The Lancet Regional Health - Americas** (2026).
- **Cardiopulmonary outcomes and Household Air Pollution (CHAP) - Social Network Study** Peru
  - Conducted the first round of data analysis examining how social network structure, tie strength, and aspirational ties relate to the adoption and sustained use of liquefied petroleum gas (LPG) stoves in rural Peru; discussed the analyses with Kendra Williams.
  - Built harmonized behavioral and network databases integrating household, exposure, and relational data for multilevel modeling, using random forest to identify predictors of LPG adoption and diffusion (n=151).
  - Prepared R Markdown analysis code and an HTML report documenting the analyses.
  - **Output**: Manuscript in preparation titled "Social Networks and Clean Fuel Use Behaviours in Rural Peru: a Social Network Analysis of the CHAP Trial."
- **HAPIN - Eye and Nutrition Substudy** Puno, Peru
  - Investigated the association between UV exposure, vitamin A/omega-3 deficiency, and ocular diseases (cataract, pterygium) among rural women in Peru (n=798).
  - Designed integrated data structures combining ophthalmologic exams, nutritional biomarkers, and environmental exposure records.
  - **Output:** Co-authored a poster presented by M. Wells at **Women in Ophthalmology Summer Symposium 2025**; eye-health manuscript submitted.
- **Global Respiratory Health Research (Chiesi)** Nepal & Peru
  - Designed and supported data collection for respiratory health assessments to expand spirometry/oscillometry screening among brick kiln workers across Nepal and Peru.
  - Designed and maintained REDCap databases to integrate multilevel data, including household, facility capacity, and individual clinical forms.
  - Performed data management, quality control, and visualization; collaborated with local teams on REDCap workflows and data reporting.
  - **Output:** Database infrastructure adopted for Chiesi Global Lung Health Program (2024–2026).
- **Global Excellence in COPD Outcomes Study (GECo) - Chronic Bronchitis Prevalence and Risk Factor Modeling** Nepal & Peru & Uganda
  - Investigated the modifiable risk factors of COPD and chronic bronchitis using population-based data from three countries among diverse LMIC populations (n = 9,664).
  - Conducted logistic regression and machine-learning analyses (random forest) using harmonized survey and spirometry data; replicated analyses, validated models through cross-validation and sensitivity testing, and prepared R Markdown outputs for the manuscript's supplementary materials.
  - Contributed analyses of associations between chronic bronchitis, lung function, and clinical outcomes.
  - Highlighted the relevance of socioeconomic disparities to COPD prevention and control.
  - **Output:** Co-author; Published in **European Respiratory Journal**; Co-authored an abstract presented by N. Robertson at **European Respiratory Society Congress 2025**.
- **GECo - Population Attributable and Impact Fractions of COPD Risk Factors** Nepal & Peru & Uganda
  - Quantified the contribution of behavioral, environmental, and clinical risk factors to COPD burden using Population Attributable Fractions (PAFs) and Potential Impact Fractions (PIFs) ($n = 10{,}709$).
  - Built integrated databases and a suite of custom R functions to standardize variable mapping, automate multilevel data linkage, and implement reproducible meta-analytic workflows across sites.
  - Modeled the combined effect of a simultaneous 25% reduction in multiple risk factors using PIFs, incorporating a causal pathway approach to account for risk factor interactions.
  - **Output:** Co-author; accepted by **The Lancet Global Health**.
- **GECo - Optimizing COPD Screening Using the SGRQ** Nepal & Peru & Uganda
  - Evaluated the St. George's Respiratory Questionnaire (SGRQ) as a population screening tool for COPD, using spirometry-defined COPD as the reference standard.

- Generated simulations for ROC and AUC analysis to evaluate the performance of SGRQ and PEF as screening tools, using the Youden index to select thresholds subject to sensitivity ≥90% in R.
- Developed an optimized PDF text extraction algorithm to address 30% missing data in a large, heterogeneous cohort of adults from Nepal, Peru, and Uganda (n=10,709).
- Evaluated a combined SGRQ–PEF screening rule that achieved 91% sensitivity and 98% negative predictive value for spirometry-defined COPD.
- Presented the poster at the American Thoracic Society International Conference 2025.
- **Output:** Co-author; Published in **American Journal of Respiratory and Critical Care Medicine** (2025).

- **HAPIN - Neurodevelopmental Outcomes in Preschool-Age Children** Puno, Peru
  - Conducted E-R analyses examining associations between prenatal and postnatal $PM_{2.5}$ and CO exposures and Bayley-III cognitive, motor, and language scores using HAPIN trial data ($n \approx 301$).
  - Implemented modified intention-to-treat models.
  - Built integrated databases and a suite of custom R functions to standardize variable mapping and automate multilevel data linkage. Constructed a socioeconomic status (SES) index using HAPIN data from Guatemala, India, Peru, and Rwanda through reproducible analytical workflows, applying iterative imputation to missing socioeconomic variables and factor analysis for mixed data.
  - **Output:** Co-author; Published in **Environmental Health Perspectives** (2025).

## WORK EXPERIENCE

- **Johns Hopkins University** Baltimore, MD
  *Senior Research Data Analyst, William Checkley: Professor of Medicine, International Health, and Biostatistics*
  Sep 2023 – May 3, 2026
  - **Global Environmental Health Analytics**: Contributed data analysis across 9 multinational studies. Managed, harmonized, and analyzed over 100 million data points spanning environmental exposure, clinical health, sensor data, questionnaire data, biomarker data, and social datasets.
  - **Pipeline Development**: Designed reproducible R/SQL workflows and internal R packages to automate data integration, cleaning, and calibration for large multi-country trials, improving reproducibility and documentation and increasing QC efficiency by over 60%.
  - **Statistical Methods & Modeling**: Applied advanced methods including linear mixed models, functional data analysis, Bayesian hierarchical modeling, and causal DAGs for multilevel environmental and health data.
  - **Data Architecture & REDCap Integration**: Built multi-tier REDCap systems linking household, individual, and facility-level data for Chiesi Foundation projects; trained local researchers on REDCap quality control and follow-up tracking.
  - **Visualization & Automation**: Automated visualization and reporting across multi-country studies, producing statistical reports, exposure maps, and interactive dashboards (ggplot2, leaflet, Shiny) supporting WHO-aligned dissemination and reproducibility.
  - **Mentorship & Collaboration**: Helped coordinate analysis meetings between JHU, UC-Berkeley, Harvard, and local field teams; provided analytical guidance to junior data analysts and contributed to manuscript methodology sections.
  - **Publications**: Contributed to four published peer-reviewed articles and two accepted articles.
- **Qiangyuan Sporting Goods Co.** Zhuhai, China
  *Data Scientist, Full-time* Jan 2021 - Sep 2021
  - **Data Infrastructure & Quality Control**: Improved data integrity across three production divisions by mapping process workflows, conducting staff interviews, and implementing Python- and R-based validation scripts to identify and correct inconsistencies in large-scale manufacturing datasets.
  - **Predictive Analytics & Forecasting**: Applied statistical modeling and machine-learning algorithms to uncover material-inventory dynamics and build demand-forecasting models, reducing excess inventory and storage costs by approximately 20%.
  - **Database Engineering**: Designed and administered SQL databases integrating procurement, production, and sales data—establishing the company's first centralized analytics platform to support real-time data querying and decision-making.
  - **Process Automation**: Streamlined manual Excel workflows into automated Python/R pipelines, generating monthly financial and procurement summaries and increasing reporting efficiency by over 80%.
- **Shandong University, Shandong Big Data Research Association** Jinan, China
  *Research Assistant, Full-time, Prof. Yufeng Shi: Statistics & Finance* Jan 2021 - May 2021
  - **Research Coordination**: Led a 12-member team through project presentations and methodological reviews, managing timelines, revision workflows, and collaborative analysis across multiple research groups.

- **Quantitative Research & Publication**: Assisted in the preparation of the second edition of *Quantitative Investment*, conducting data validation, literature synthesis, and empirical case updates on portfolio optimization and statistical trading models.

## TEACHING EXPERIENCE

- **Department of Applied Mathematics and Statistics, Johns Hopkins University** Baltimore, MD
  Teaching Assistant: Introduction to Data Science Jan 2022 - May 2022
  - Assisted in lab instruction, assignment design, and project grading.
  - Supported students in statistical learning methods (regression, SVMs, PCA) and Python implementation.
- **Department of Applied Mathematics and Statistics, Johns Hopkins University** Baltimore, MD
  Teaching Assistant: Computing for Applied Mathematics Sep 2022 - Dec 2022
  - Led weekly teaching sessions, held office hours, and designed weekly quizzes for ∼30 graduate students covering Python programming, numerical methods, and data manipulation for applied mathematics.
  - Provided support in debugging, optimizing code, and interpreting computational results while grading weekly assignments, quizzes, and final projects with detailed feedback on both correctness and code style.

## TECHNICAL SKILLS

- GRE: Verbal 164; Quantitative 170
- Programming & Statistical Computing: R, Python, MATLAB, SAS, SQL, Java, R Markdown, LaTeX, Git, AWS, REDCap, Tableau
- Geospatial Analysis: ArcGIS, R sf/terra/raster packages, custom shapefile processing, spatial interpolation (Kriging), GIS-based exposure surface generation
- Environmental Monitoring & Field Instrumentation: PurpleAir low-cost $PM_{2.5}$ sensor co-location, deployment, and reference-grade calibration against MetOne E-Sampler monitors; ambient air-quality network design and quality control for LMIC field campaigns